

1. What is the output of the following Java fragment code?

```
int n = 6, m = 15, p = 3;
n -= m;
n *= p;
System.out.println(n);
```

- A. 3
 - B. -27
 - C. 24
 - D. 27
2. Which concept refers to extending specialized classes from generalized classes in object-oriented programming?
- A. Overloading
 - B. Data field Encapsulation
 - C. Overriding
 - D. Inheritance
3. What is the output of the following Java fragment code?

```
int[] list = new int[4];
int sum = 0;
for(int i=0; i< list.length; i++){
    list[i] = i * 3;
    sum+=list[i];
}
System.out.println(sum);
```

- A. 3
 - B. 36
 - C. 18
 - D. 6
4. Which one of the following method signatures represents argument passing by value?
- A. double args (int x, double y)
 - B. double args (T a)
 - C. T args ()
 - D. A and B
 - E. None of the above
5. What is the output of the below given Java program?

```
public class MidExam {
    String str="Mid";
    String str1;
    MidExam() {
        this("Exam");
        System.out.println(str+" ");
    }
    MidExam(String s) {
        this(s, "Java");
        str = s;
        System.out.print(str+" ");
    }
}
```

```

MidExam(String s, String str){
    str1 = str;
    str = s;
    System.out.print(str+" ");
}
public static void main(String[] args) {
    MidExam me = new MidExam();
}
}

```

- A. Java Mid Exam
 - B. Mid Exam Java
 - C. Exam Exam Exam
 - D. Mid Mid Mid
 - E. None of the above
6. What is the output of the following program? (3 Pts.)

```

public class Output {
    static int n=7;
    int m=2;
    static {
        n++;
        System.out.println("Static: "+ n);
    }
    {
        m++;
        System.out.println("Instance: "+ m);
    }
    Output() {
        m++;
        System.out.println("Constructor: "+m);
    }
    public static void main(String[] args) {
        Output out = new Output();
        System.out.println("Main: "+ out.m);

        out.show(out.m++, n);
    }
    void show(int a, int b) {
        n = a;
        m = b;
        System.out.println("Show: m = "+ m);
        System.out.println("Show: n = "+ n);
    }
}

```

7. Which one of the following polymorphism types is not implemented using method overloading?
- A. Late Binding
 - B. Static binding
 - C. Compile-time
 - D. None of the above

8. Which one of array initialization statements is correct?
- A. `int num [5] = {3, 5, 7, 12, 20};`
 - B. `char s [][] = {21, 23, 33, 45};`
 - C. `var num = {5, 6, 9, 21};`
 - D. `boolean b [] = {false, true, false, true};`

9. What is the output of the below given Java code?

```
byte ch[] = {8, 6, 10, 3, 7, 1};
ch = new byte[6];
int sum=0;
for(byte b: ch) {
    sum += b;
}
System.out.println(sum);
```

- A. 35
 - B. 0
 - C. Syntax Error
 - D. 8 6 10 3 7 1
10. Which declaration of constant is **CORRECT** in the class body?
- A. `static int NUM = 3;`
 - B. `int NUM = 4;`
 - C. `final static int NUM = 3;`
 - D. `int NUM;`

11. Consider the following code

```
void start() {
    A a = new A();
    B b = new B();
    a.s(b);
    b = null; /* Line 5 */
    a = null; /* Line 6 */
    System.out.println("start completed"); /* Line 7 */
}
```

When is the **B's** object, created in line 3, eligible for garbage collection?

- A. after line 5
 - B. after line 6
 - C. after line 7
 - D. Not certain
12. Which one of the following two-dimensional array declarations in Java is **WRONG**?
- A. `int [][] x = new int[2][3];`
 - B. `int x [][] = new int[2][];`
 - C. `int x [][] = new int[][3];`
 - D. `int [][] x = new int[2][];`
13. What is the output of the following code?

```
class MyProgram {
    String str = "Mid Exam";
    String str1= "";
```

```

        public static void main(String args[ ]) {
            for(int s = str.length( ) - 1; s >= 0; s++)
                str1 += str.toUpperCase(str.charAt(s));
            System.out.println(str1);
        }
    }

```

- A. Mid Exam
- B. MID EXAM
- C. maxE diM
- D. None of the above

14. Which one of the following **String constant** declaration statements in the **Interface** body is **Wrong**?

- A. String str = "Peace";
- B. static String str = "Peace";
- C. final String str = "Peace";
- D. None of the above

15. One of the following is false about **super()**.

- A. It must be the first statement in the constructor body
- B. Is used to call superclass's constructor from subclass's constructor
- C. It cannot be used in the user defined method.
- D. All of the above
- E. None of the above

16. A method must do either of two things with a checked exception. What are these things?

- A. (1) Ignore the exception, or (2) check the exception.
- B. (1) Handle the exception in a catch{} block, or (2) return the exception to the sender.
- C. (1) Handle the exception in a catch{} block, or (2) throw the exception to the method that called this method.
- D. (1) Handle the exception in a catch{} block, or (2) handle the exception in a try{} block,

Consider the following code and answer the below give question

```

final class Test{
    public void display(){
        System.out.print("This is a test machine");
    }
}

```

17. Which one of the following statements is **WRONG**?

- A. The method display cannot be overridden
- B. The class cannot be inherited
- C. Objects cannot be instantiated from the class
- D. The method display can be overloaded

18. What is the output of the following Java code?

```

int num[] = {2,4,6,8};
int y;
try {
    for(int i=0; i <= num.length; i++){

```

```

        y = num[i] / i;
        System.out.println("Quotient: " + y);
    }
}
catch (ArithmeticException ae) {
    System.out.println("Division By Zero");
}
catch (ArrayIndexOutOfBoundsException e) {
    System.out.println("Array Size Problem");
}
catch (Exception a) {
    System.out.println("Any Exception");
}
finally{
    System.out.println("Good Day");
}

```

- A. Division By Zero
- B. Any Exception
Good Day
- C. Array Size Problem
Any Exception
- D. Division By Zero
Any Exception
Good Day
- E. None of the above